

Complete Research Methodology

Reliability-Aware Rank Aggregation for Deployment-Oriented Predictive Maintenance Model Selection

1. Introduction and Methodological Purpose

Predictive maintenance (PdM) detects degradation or failure risk before to disruptive equipment failure by using operational data and condition-monitoring. Machine-learning and deep-learning approaches are increasingly employed for fault classification, condition monitoring and remaining useful life (RUL) prediction [1]-[5]. However, recent evaluations continue to highlight model selection, deployment sustainability, robustness, and changeable operating conditions as significant obstacles for AI-driven PdM [1], [2], [5].

Deployment suitability is not always shown by strong prediction performance alone. A model configuration may display poor robustness to sensor degradation, unreliable probability estimates, excessive computational requirements, or undesirable operational consequences while concurrently performing well on a classification or regression metric. Thus, deployment-oriented PdM model selection is treated in this research as a multi-dimensional ranking problem.

Five deployment dimensions are considered:

1. predictive quality
2. robustness
3. probability reliability
4. computational efficiency
5. operational impact

These dimensions may produce conflicting rankings. The configuration with the best predictive quality might not have the best operational impact, robustness, or efficiency. Research on rank-aggregation and multi-criteria decision-making offers well-established methods for integrating conflicting data [12]-[16]. However, a further problem arises when the empirical evidence supporting the criterion rankings is itself differently reproducible.

For example, repeated experiments may show that one criterion produces a highly stable ordering of candidate configurations, while another criterion produces frequent rank changes. Despite this disparity in empirical reproducibility, equal-weight aggregation provides each criterion rankings the same influence.

Therefore, the methodological problem addressed by this research is: How should conflicting deployment rankings be aggregated when the empirical evidence supporting those rankings is not equally reliable?

Borda scoring, Kendall and Spearman rank agreement, weighted aggregation, Pareto analysis and regret analysis are established techniques, and prior work has already investigated ensemble and robust rank aggregation [15], [16].

Hence, the contribution tested here is to estimate the empirical reproducibility of deployment-criterion rankings from repeated predictive-maintenance evaluation evidence and use that stability as reliability information within deployment-oriented rank aggregation.

Thus, the overall logic is:

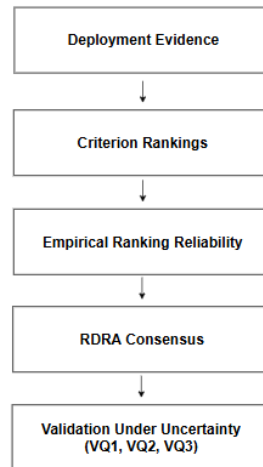


Figure 1. Overall Research Methodology Logic

This structure directly links the research gap with the proposed method and its validation.

2. Research Objective and Research Questions

2.1 Research Objective

The principal objective is:

To develop and empirically validate a reliability-aware rank-aggregation methodology for deployment-oriented predictive-maintenance model selection that accounts for differences in the stability of evaluation evidence and produces a defensible consensus ranking under evaluation uncertainty.

Consequently, the research moves beyond the question:

Which predictive-maintenance model achieves the strongest benchmark performance?

towards

Which deployment recommendation is most defensibly supported when multiple deployment criteria conflict and their supporting evidence is not equally reliable?

2.2 Main Research Question

The Main Research Question is:

How can reliability-aware rank aggregation produce a robust and defensible consensus ranking of predictive-maintenance model configurations when deployment criteria provide conflicting and differently stable evaluation evidence?

Three validation questions examine complementary properties required to answer the Main Research Question.

VQ1 - Criterion Uncertainty: Consistency

Does reliability-aware aggregation improve consensus-ranking consistency when the available deployment criteria vary?

A useful consensus ranking should not change excessively when reasonable subsets of the deployment criteria are considered.

VQ2 - Data-Quality Uncertainty: Responsiveness

Does the consensus ranking respond appropriately when meaningful changes occur in the underlying deployment evidence due to sensor degradation?

Stability alone is insufficient because a meaningless fixed ordering could also be stable. VQ2 therefore examines whether ranking changes correspond to meaningful changes in the underlying evidence.

VQ3 - Preference Uncertainty: Decision Usefulness

How useful is the consensus ranking for downstream deployment decisions when deployment priorities vary?

VQ3 evaluates candidate shortlists and ranking usefulness under computationally sampled preference-weight scenarios.

A central construct distinction is maintained throughout the research:

$$\textit{empirical reliability weight} \neq \textit{preference weight}$$

Reliability represents the reproducibility of criterion-ranking evidence, whereas preference represents the relative importance assigned to a criterion within a decision context.

3. Research Design and Method Selection

3.1 Overall Research Design

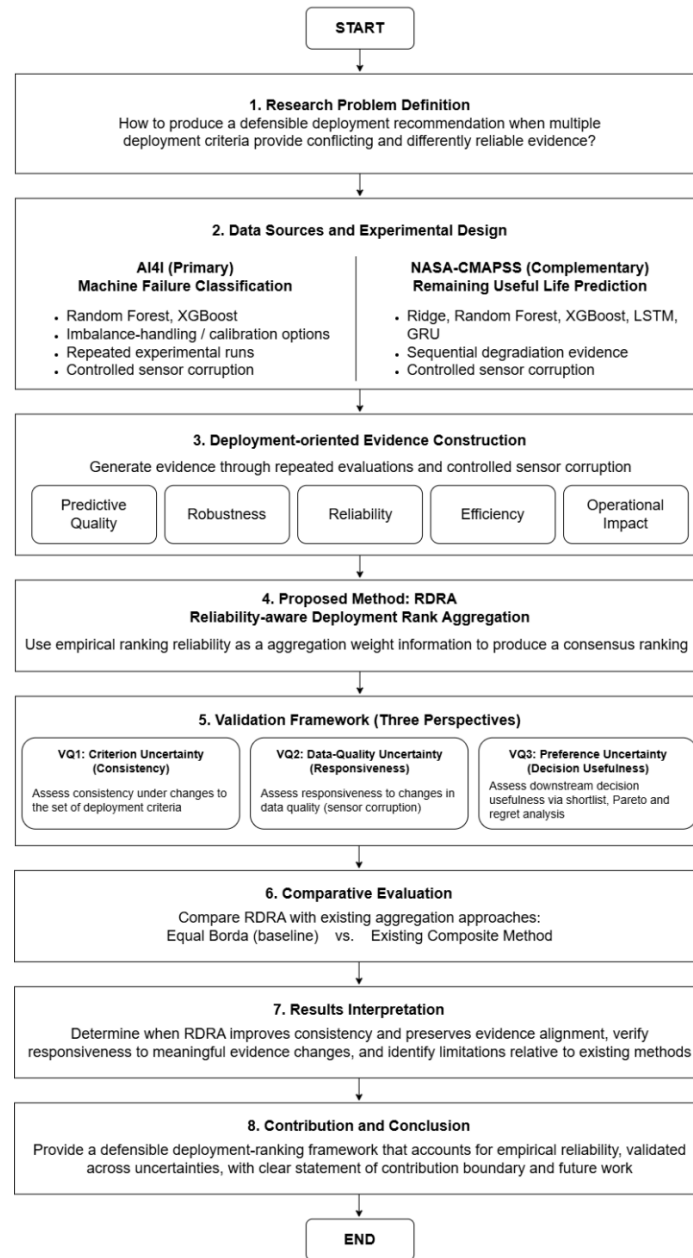


Figure 2. Overall research methodology for the development and validation of Reliability-Aware Deployment Rank Aggregation.

Figure 2 summarises the overall research design and shows how the research problem is translated into experimental evaluation and validation. AI4I is used as the primary environment for RDRA development, while C-MAPSS provides complementary cross-task evidence. The framework links deployment-oriented evidence construction with the proposed RDRA method,

three validation perspectives, comparative evaluation and final interpretation, ensuring that each stage of the methodology directly contributes to answering the Main Research Question.

This research adopts a quantitative computational experimental design combining:

- comparative predictive modelling
- repeated experimental evaluation
- controlled sensor-data perturbation
- multi-dimensional deployment evaluation
- criterion-level ranking
- empirical ranking-reliability estimation
- rank aggregation
- statistical comparison
- sensitivity analysis
- downstream decision analysis

3.2 Alternative Research Methods Considered

Qualitative Industrial Case Study

A real industrial case study could provide high contextual realism, direct stakeholder involvement and exposure to actual organisational constraints. Nevertheless, it would offer less experimental control over repeated ranking evidence, criterion perturbations and sensor degradation. Proprietary data could also create access and confidentiality constraints.

Thus, a case study is more suitable for future external validation than for the primary experimental investigation of RDRA.

Survey or Interview Research

Surveys or interviews could determine how practitioners value predictive quality, robustness, efficiency or maintenance risk.

Such evidence would be useful for modelling criterion preference, but it would not estimate how reproducibly experimental evidence ranks competing models. Hence, it is not sufficient for the central reliability construct investigated by RDRA.

Pure Simulation

A fully synthetic simulation could provide precise control over ranking noise and known ground truth. Its main limitation is weaker connection to actual PdM model behaviour.

Instead, to maintain experimental control while maintaining a connection to practical prediction tasks, this research integrates benchmark datasets with controlled perturbations.

Performance-Only Benchmarking

A conventional benchmark based only on F1, ROC-AUC, RMSE or MAE would be simple and reproducible, but it would not address conflicting deployment criteria or ranking reliability.

Conventional MCDM

MCDM provides mature methods for evaluating alternatives across multiple criteria [12], [13]. However, conventional weights commonly represent importance or preference rather than empirical ranking reproducibility.

Equal Borda Aggregation

Equal Borda is retained as the principal rank-based baseline because it is simple, transparent and scale independent. Its defining assumption is:

$$w_j = \frac{1}{m}$$

which means that all criterion rankings receive equal influence regardless of their empirical stability.

3.3 Justification for the Selected Method

The selected quantitative computational experimental design is appropriate because it directly supports the type of evidence required to solve the Main Research Question. First, the research examines measurable ranking behaviour, including ranking consistency, responsiveness and agreement, which makes a quantitative design more suitable than a purely qualitative approach. Second, repeated experimental evaluations allow criterion-ranking reliability to be estimated empirically from observed variation across repetitions rather than being assigned subjectively. Third, controlled perturbations make it possible to systematically vary criterion availability and sensor-data quality while keeping the candidate configurations, evaluation procedures and comparison conditions consistent. This improves internal validity and enables fair comparison between RDRA and the baseline aggregation methods.

Furthermore, the design facilitates falsifiability and reproducibility. The analysis may be reproduced and independently verified due to documented preprocessing, repeatable computational notebooks, defined experimental techniques, and public benchmark datasets. Importantly, RDRA is not assumed to be superior simply because it is the proposed method. Its additional complexity is justified only if predefined validation analyses demonstrate meaningful advantages or useful behaviour relative to simpler alternatives such as Equal Borda and the Existing Composite method. This makes the research design suitable for producing valid, reliable

and defensible evidence about whether reliability-aware aggregation improves deployment-oriented predictive-maintenance model selection.

4. Datasets and Experimental Context

4.1 AI4I 2020

The AI4I 2020 Predictive Maintenance Dataset is the primary environment for RDRA development and reliability-aware validation [6].

The task is machine-failure classification. Candidate configurations include Random Forest and XGBoost, together with imbalance-handling and probability-calibration alternatives. Repeated experimental runs provide the evidence required to estimate criterion-ranking stability, while controlled sensor corruption supports evaluation under data-quality uncertainty.

AI4I is used for:

- five-dimensional deployment evaluation
- criterion-level ranking
- reliability estimation
- RDRA development
- VQ1, VQ2, and VQ3

4.2 NASA C-MAPSS

NASA C-MAPSS FD001 provides complementary RUL-prediction evidence. RUL prediction is a well-established machinery-prognostics problem [4].

The candidate methods include:

- Ridge
- Random Forest
- XGBoost
- LSTM
- GRU

Unlike AI4I's tabular classification setting, C-MAPSS provides sequential degradation evidence. Controlled sensor corruption is applied and the resulting rankings are recomputed to investigate deployment-rank responsiveness in a different task.

4.3 Scope of Cross-Task Evaluation

AI4I provides the primary and full RDRA evaluation environment, because its repeated experimental evidence supports criterion-level ranking, empirical reliability estimation, reliability-weighted aggregation, and validation under the three perspectives of criterion uncertainty, data-quality uncertainty and preference uncertainty.

In contrast, C-MAPSS is used as a complementary cross-task evaluation in a different predictive-maintenance setting, namely Remaining Useful Life prediction. The C-MAPSS analysis examines whether deployment rankings respond meaningfully when sequential sensor data are subjected to controlled degradation, but it does not estimate criterion reliability weights from repeated experiments.

Therefore, C-MAPSS is not treated as a complete replication of the AI4I RDRA procedure. Instead, its role is to provide additional evidence that the broader deployment-ranking framework can be examined beyond the primary machine-failure classification setting, while avoiding an unsupported claim that the complete reliability-aware method has been independently replicated across both datasets.

5. Deployment-Oriented Evidence Construction

5.1 Predictive Quality

Predictive quality captures how effectively each candidate performs its primary predictive task and therefore provides the foundational evidence for model comparison.

For AI4I, the predictive-quality dimension combines normalized F1-score and PR-AUC. These measures are used because the machine-failure task is imbalanced, making them more informative than relying on accuracy alone. F1-score balances precision and recall, while PR-AUC summarises performance across decision thresholds and is particularly relevant when the positive failure class is comparatively rare.

For C-MAPSS, predictive quality is evaluated using RUL-prediction error measures such as RMSE and MAE. These measures quantify the difference between predicted and actual remaining useful life and provide an appropriate basis for evaluating regression-based prognostic performance.

5.2 Robustness

Robustness measures how well model performance is preserved when sensor data is degraded. This dimension is crucial because predictive-maintenance systems may operate under noisy, incomplete or faulty sensing conditions, and robustness to such conditions remains an important challenge in industrial fault diagnosis [1], [7].

The AI4I corruption protocol includes four families:

- Gaussian noise
- Missing Completely At Random
- outliers
- static sensor offset

Multiple severity levels are applied within each family to examine whether candidate performance degrades gradually or disproportionately as data quality worsens. Including several corruption types reduces the risk of defining robustness from only one artificial fault condition and offers broader evidence about model behaviour under sensor uncertainty.

5.3 Probability Reliability

Probability reliability is evaluated separately from predictive discrimination because a model can classify observations accurately while still producing poorly calibrated probability estimates [8]-[10]. Reliable probability estimates are important in deployment contexts where predicted risk may influence maintenance prioritisation or operational decisions.

The AI4I probability-reliability dimension combines cost-normalised:

- Brier score
- Expected Calibration Error

The Brier score evaluates the accuracy of probabilistic predictions, while Expected Calibration Error measures the agreement between predicted confidence and observed outcomes. Calibration configurations include uncalibrated and calibrated variants so that the analysis considers not only predictive discrimination but also the reliability of the confidence information associated with model predictions.

5.4 Computational Efficiency

Computational efficiency captures whether a candidate can provide suitable predictive behaviour without excessive deployment overhead. The efficiency dimension includes:

- inference latency
- model size

Inference latency represents the time required to generate predictions, while model size provides an indication of storage and deployment-resource requirements. Including these indicators reflects the practical trade-off that a small improvement in predictive quality may not justify substantially greater computational cost, particularly in resource-constrained or near-real-time monitoring environments.

5.5 Operational Impact

Operational impact represents the practical consequences of prediction errors. In predictive maintenance, false negatives and false positives may have substantially different implications. Missing an emerging fault may increase the risk of unplanned failure, whereas a false alarm may result in unnecessary inspection or maintenance activity. Thus, PdM research has recognised the importance of non-equivalent failure costs [11].

The operational penalties used in this study are analytical scenario assumptions rather than measured industrial financial costs. Their goal is to represent asymmetric decision consequences consistently across candidate configurations and support comparative analysis without claiming that the assigned values correspond to actual industrial monetary losses.

5.6 Criterion-Level Ranking

After the deployment evidence has been constructed, each dimension is converted into a criterion-level ranking.

For candidate i and criterion j :

$$r_{ij}$$

denotes the candidate's rank, where:

$$r_{ij} = 1,$$

represents the most desirable candidate for criterion j .

Average ranking is used when candidates are tied, avoiding arbitrary ordering between alternatives with equivalent evidence.

Rank transformation is appropriate because the deployment dimensions are expressed using heterogeneous measures with different scales and units. For example, predictive scores, calibration errors and inference latency cannot be meaningfully added in their raw form. Converting each dimension into an ordinal ranking provides a common representation that can subsequently be aggregated using Borda-based methods while preserving the relative ordering produced by each deployment criterion.

6. Proposed Reliability-Aware Deployment Rank Aggregation

6.1 Methodological Rationale

Established rank-aggregation research demonstrates that multiple rankings can be combined into a consensus [14]-[16]. RDRA adapts this general principle by using empirical criterion-ranking reproducibility as reliability information in the aggregation process.

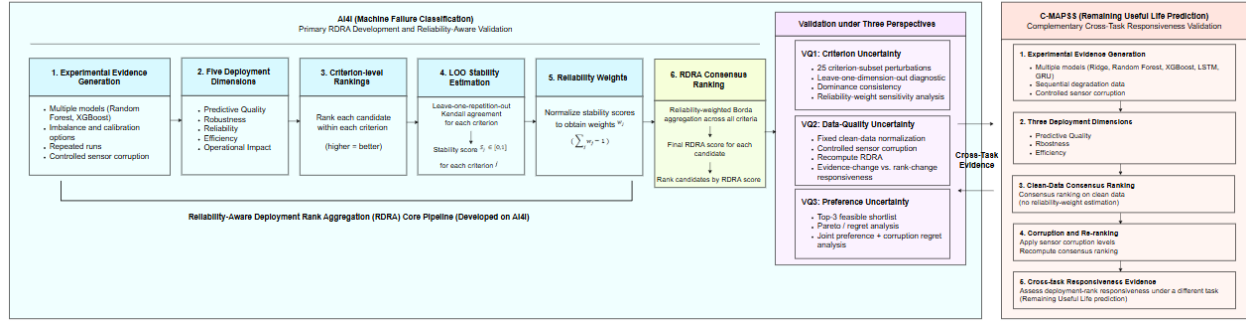


Figure 3. Detailed experimental framework for RDRA development and validation.

Figure 3 provides detailed experimental architecture for research. It shows how repeated AI4I evidence is transformed through the five deployment dimensions, criterion-level rankings, Leave-One-Repetition-Out stability estimation and reliability weighting before producing the RDRA consensus ranking. The figure also separates the three validation perspectives: criterion uncertainty, data-quality uncertainty and preference uncertainty, and clarifies that C-MAPSS is used only as complementary cross-task responsiveness evidence. This structure is important because it makes the relationship between method development, validation and cross-task evaluation explicit while preventing the C-MAPSS analysis from being interpreted as a full reliability-weighted replication of RDRA.

6.2 RDRA Algorithm

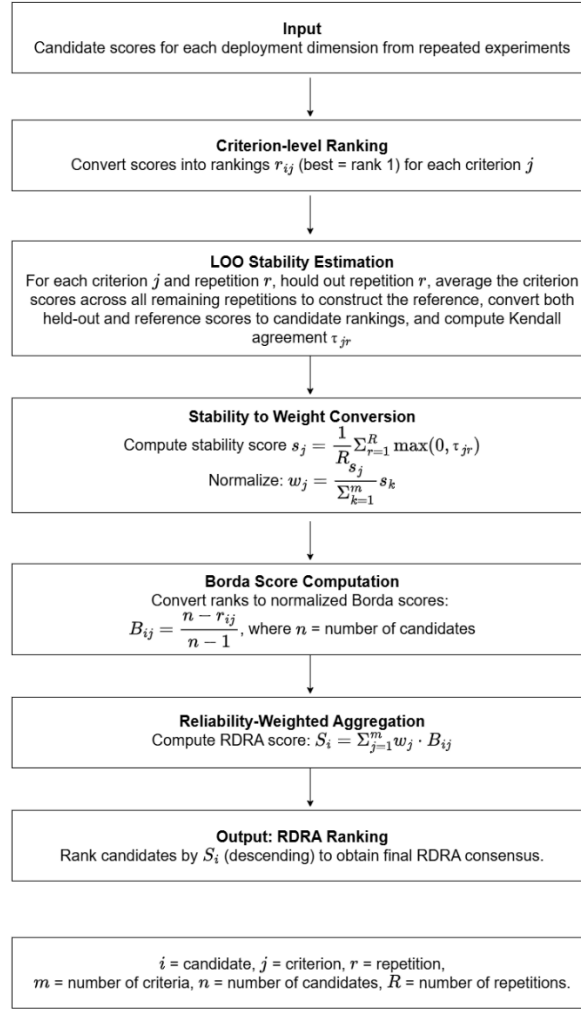


Figure 4. Algorithmic workflow of Reliability-Aware Deployment Rank Aggregation.

Figure 4 specifies the internal calculation performed by RDRA. Repeated criterion scores are first converted into candidate rankings, after which Leave-One-Repetition-Out Kendall agreement is used to estimate the reproducibility of each criterion ranking. These stability estimates are normalised into reliability weights and combined with normalised Borda scores to produce the final RDRA consensus. The figure is included to make the proposed method transparent and reproducible by clearly identifying its inputs, intermediate calculations and final output, while distinguishing the empirical reliability-estimation step from established Borda-based rank aggregation.

6.3 Criterion-Level Ranking

Candidate deployment scores are first converted into criterion-level rankings. For candidate i and criterion j ,

$$r_{ij}$$

denotes the candidate's rank, where rank 1 represents the most desirable candidate under that criterion. This transformation is necessary because the deployment dimensions are measured using different scales and units. Converting the evidence to rankings provides a common ordinal representation that preserves the relative ordering of candidates without directly combining incompatible raw measures. Average ranks are retained where ties occur so that equivalent evidence does not produce an arbitrary ordering.

6.4 Leave-One-Repetition-Out Stability Estimation

The empirical reliability of each criterion ranking is estimated using a Leave-One-Repetition-Out (LOO) procedure. For every criterion j and repetition r :

1. repetition r is held out
2. each candidate's criterion score is averaged across the remaining repetitions
3. these averages form the reference evidence
4. held-out scores and reference scores are independently converted into candidate rankings
5. Kendall agreement τ_{jr} is computed between the rankings

Thus, the stability score is:

$$s_j = \frac{1}{R} \sum_{r=1}^R \max(0, \tau_{jr})$$

where R is the number of experimental repetitions.

This procedure measures how consistently criterion j orders the candidates when individual repetitions are removed. A high s_j indicates that the criterion produces reproducible ordering, whereas a lower value indicates greater sensitivity to experimental variation. Negative Kendall agreement is truncated at zero so that inverse or contradictory ranking behaviour cannot generate a negative reliability weight.

This step is central to RDRA because the reliability information is estimated from repeated experimental evidence rather than assigned subjectively.

6.5 Reliability Weights

The criterion stability estimates are converted into normalised reliability weights:

$$w_j = \frac{s_j}{\sum_{k=1}^m s_k}$$

subject to:

$$0 \leq w_j \leq 1$$

and:

$$\sum_{j=1}^m w_j = 1$$

Normalisation places all criterion weights on a common scale and ensures that their total contribution to the final aggregation equals one. Criteria whose rankings are more reproducible across repetitions receive greater influence, while less stable criteria receive comparatively lower influence. Furthermore, w_j represents empirical ranking reliability, not stakeholder preference or criterion importance. This differences avoid the reliability mechanism from being interpreted as a subjective weighting scheme.

6.6 Normalised Borda Scores

For n candidate configurations, each criterion rank is converted into a normalised Borda score:

$$B_{ij} = \frac{n - r_{ij}}{n - 1}$$

The highest-ranked candidate therefore receives:

$$B_{ij} = 1$$

while the lowest-ranked candidate receives:

$$B_{ij} = 0$$

Normalised Borda scoring converts ordinal ranks into a common bounded scale while preserving their relative ordering. This enables the criterion rankings suitable for weighted aggregation and ensures that criteria with different original measurement scales remain directly comparable in the consensus calculation.

6.7 Reliability-Weighted Aggregation

The final RDRA score for candidate i is calculated as:

$$S_i = \sum_{j=1}^m w_j B_{ij},$$

where w_j is the empirical reliability weight for criterion j , and B_{ij} is the candidate's normalised Borda score under that criterion.

Candidates are ranked by S_i in descending order, so a higher RDRA score represents stronger overall support across the deployment criteria after accounting for differences in empirical ranking reliability.

Therefore, the core RDRA procedure can therefore be summarised as:

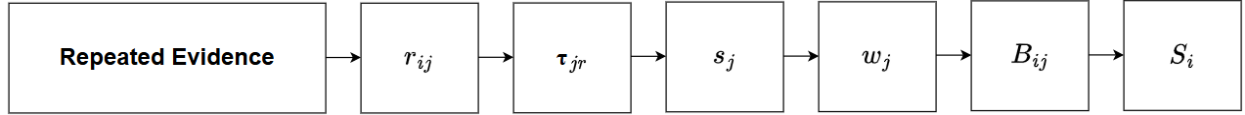


Figure 5. Core RDRA procedure

This sequence makes the role of the proposed reliability mechanism explicit: repeated experimental evidence is first used to estimate ranking reproducibility, and that information then determines how strongly each criterion contributes to the final consensus.

6.8 Reliability-Differentiation Refinement

The core RDRA formulation applies reliability weights linearly. A parameterised extension is additionally evaluated to determine whether the **strength of reliability differentiation** affects ranking consistency:

$$w_j(\gamma) = \frac{s_j^\gamma}{\sum_{k=1}^m s_k^\gamma}.$$

When $\gamma = 0$, the weights approach equal weighting, corresponding to the Equal Borda limit.

When $\gamma = 1$, the original linear RDRA weighting is recovered.

For $\gamma > 1$, differences between more stable and less stable criteria are amplified, giving relatively greater influence to criteria whose rankings are more reproducible.

To avoid selecting γ using the same evidence later used for validation, the 25 VQ1 criterion-subset perturbations are deterministically divided into

- 15 development subsets
- 10 held-out subsets

Candidate γ values are evaluated only on the development subset using a composite objective based on Kendall agreement, Top-3 overlap and inverse normalised rank displacement. The selected value is then locked before evaluation on the held-out VQ1 subset and subsequent VQ2 and VQ3 analyses. This development/validation separation reduces parameter-selection bias and provides a more defensible assessment of the refined weighting strategy.

The refinement does not imply that the original RDRA formulation is incorrect. Instead, it evaluates whether the same reliability-aware principle benefits from stronger or weaker differentiation between empirical stability estimates. If the strongest development result occurs at the upper or lower boundary of the tested γ range, additional development-only boundary sensitivity analysis should be considered before describing that value as globally optimal.

7. Validation and Statistical Analysis

7.1 Criteria for a Defensible Ranking

The validation framework evaluates four complementary properties:

- evidence alignment
- reliability awareness
- consistency
- responsiveness

Evidence alignment requires the consensus ranking to remain grounded in meaningful deployment evidence rather than arbitrary ordering. Reliability awareness evaluates whether differences in empirical criterion stability are represented during aggregation. Consistency evaluates whether the ranking remains reasonably stable when the available deployment criteria vary, while responsiveness evaluates whether the ranking changes when the underlying evidence changes meaningfully.

These properties are considered jointly because stability alone does not establish ranking quality. A ranking that never changes could appear highly consistent while being unrelated to the deployment evidence. Thus, the validation framework evaluates whether RDRA achieves an appropriate balance between robustness to minor analytical variation and sensitivity to meaningful changes in the evidence.

7.2 VQ1 - Criterion Uncertainty

VQ1 evaluates the consistency of the consensus ranking when the set of available deployment criteria changes.

With five deployment dimensions:

$$\binom{5}{2} = 10,$$

$$\binom{5}{3} = 10,$$

and:

$$\binom{5}{4} = 5.$$

Therefore:

$$N = 10 + 10 + 5 = 25$$

criterion-subset perturbations are evaluated.

The complete five-dimensional ranking acts as the corresponding reference ranking for each aggregation method. The five-dimensional case itself is not counted as a perturbation.

Three aggregation methods are compared:

- Equal Borda
- Existing Composite
- RDRA

Using the same criterion subsets for all methods ensures a matched comparison and allows differences in consistency to be attributed to the aggregation approach rather than to different perturbation conditions.

VQ1 Metrics

The following complementary measures are used:

- Kendall agreement
- Spearman agreement
- Top-3 overlap
- mean absolute rank displacement
- tie-aware winner survival
- Top-1 set overlap.

Kendall agreement assesses preservation of pairwise ordering, while Spearman agreement captures overall monotonic similarity between rankings. Top-3 overlap evaluates whether the practical deployment shortlist is preserved, which is useful because operational decisions may focus on a small number of leading candidates rather than the complete ranking.

Mean absolute rank displacement is defined as:

$$D = \frac{1}{n} \sum_{i=1}^n |r_i^{perturbed} - r_i^{reference}|$$

Lower values indicate stronger ranking consistency because candidates move fewer positions relative to the full-criteria reference.

Tie-aware winner survival and Top-1 set overlap are included so that tied candidates are not forced into an arbitrary single-winner interpretation.

Additional VQ1 diagnostics include:

- leave-one-dimension-out analysis
- dominance consistency
- reliability-weight sensitivity

These diagnostics complement the primary 25-subset analysis by examining whether the conclusions remain coherent under simpler criterion-removal and weight-perturbation conditions.

7.3 VQ1 Statistical Validation

Because all aggregation methods are evaluated under the same 25 criterion-subset perturbations, the resulting observations are paired.

For a metric where higher values indicate better consistency:

$$d_i = M_{RDRA,i} - M_{Borda,i}.$$

For mean absolute rank displacement, the direction is reversed so that a positive difference consistently represents improvement in favour of RDRA.

The primary statistical analysis reports:

- mean paired difference
- 95% confidence interval
- paired t -test
- Cohen's d_z

- Holm multiple-comparison correction [19].

The paired t -test is appropriate because the same perturbation conditions are evaluated by both methods, allowing the analysis to focus on within-condition differences rather than independent groups. Cohen's d_z is reported to quantify the magnitude of the paired effect, while confidence intervals communicate the uncertainty surrounding the estimated improvement.

Because several ranking metrics are tested, Holm correction is applied to control the family-wise error rate without using the more conservative adjustment associated with a simple Bonferroni procedure [19].

The Shapiro-Wilk test is used to examine the distribution of paired differences. Where normality is questionable, the Wilcoxon signed-rank test is additionally reported as a non-parametric robustness check.

Therefore, the interpretation is considered

*direction + effect size + confidence interval + adjusted significance
+ practical ranking meaning*

rather than depend on p -values alone.

7.4 VQ2 - Data-Quality Uncertainty

VQ2 evaluates whether the consensus ranking responds appropriately when the underlying deployment evidence changes because of sensor-data degradation.

Four corruption families and four severity levels produce:

$$4 \times 4 = 16$$

AI4I data-quality scenarios.

The corruption families are:

- Gaussian noise
- Missing Completely At Random
- Outliers
- static sensor offset

The primary VQ2 analysis uses fixed clean-data normalisation. Clean-data scaling parameters are retained when corrupted scenarios are evaluated so that changes in the evidence are measured relative to the same reference scale. This reduces the risk that scenario-specific re-scaling masks the magnitude of degradation.

For each scenario:

ΔE = magnitude of deployment-evidence change,

and:

ΔR = magnitude of consensus-ranking change.

Responsiveness is assessed using:

$$\rho = \text{Corr}_S(\Delta, E\Delta R),$$

where Corr_S denotes Spearman rank correlation.

A stronger positive correlation indicates that larger evidence changes tend to be associated with larger ranking changes. This is desirable because the consensus should remain relatively stable under minor evidence variation but respond when the deployment evidence changes substantially.

Adaptive normalisation is retained as a sensitivity analysis to determine whether the responsiveness conclusions depend on the normalisation strategy.

7.5 VQ2 Statistical Validation

To compare responsiveness directly between RDRA and Equal Borda:

$$\Delta\rho = \rho_{RDRA} - \rho_{Borda}$$

Bootstrap resampling is used to quantify uncertainty around this difference [18].

For:

$$B = 10,000$$

matched bootstrap samples:

$$\Delta\rho^{(b)} = \rho_{RDRA}^{(b)} - \rho_{Borda}^{(b)}$$

A percentile 95% bootstrap confidence interval is then obtained from the bootstrap distribution.

This direct comparison is preferable to inferring superiority merely because one individual correlation is statistically significant while another is not. The bootstrap interval instead evaluates whether the observed difference in responsiveness is consistently supported across resampled corruption scenarios.

7.6 VQ3 - Preference Uncertainty

VQ3 examines downstream decision usefulness rather than conducting a separate inferential statistical-validation stage.

The analysis includes:

- feasible Top-3 shortlists
- Pareto analysis
- sampled deployment-preference weights
- regret analysis
- joint preference-plus-corruption regret analysis.

For candidate i under preference scenario q :

$$U_{iq}$$

denotes the scenario-specific utility.

The best achievable utility is:

$$U_q^* = \max_i U_{iq}$$

and candidate regret is:

$$Regret_{iq} = U_q^* - U_{iq}$$

Regret measures the opportunity loss associated with selecting a candidate instead of the best available candidate under a particular sampled preference scenario.

Hence, the VQ3 analysis evaluates whether the ranking methods produce useful candidate shortlists under varying deployment priorities. Conclusions remain conditional on the candidate set, sampled preference distribution, feasibility constraints, corruption scenarios and operational assumptions.

7.7 C-MAPSS Complementary Responsiveness Analysis

C-MAPSS provides complementary cross-task evidence through the following process, as shown in Figure 6.

Ranking agreement between clean and corrupted scenarios is evaluated using measures such as Kendall agreement.

The purpose of this analysis is to determine whether deployment-ranking behaviour can also be examined under sequential degradation evidence in a different PdM task. Because C-MAPSS does not include the repeated criterion-level reliability estimation required for the full RDRA procedure, this evidence is treated as complementary cross-task responsiveness analysis rather than a complete RDRA replication.

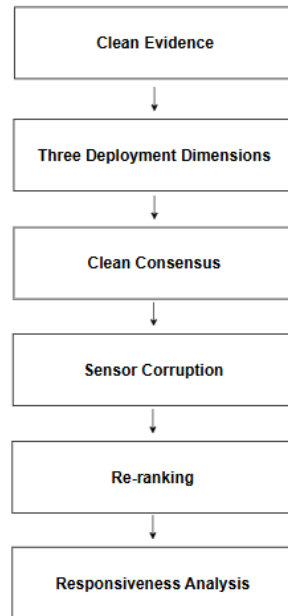


Figure 6. C-MAPSS Complementary Responsiveness Analysis Process

7.8 Sensitivity Analysis

Sensitivity analysis evaluates whether the conclusions depend excessively on one analytical assumption, parameter choice or normalisation strategy.

The checks include:

- reliability-weight perturbation
- leave-one-dimension-out diagnostics
- criterion-subset perturbation
- γ sensitivity
- fixed versus adaptive normalization
- development versus held-out evaluation
- bootstrap uncertainty
- comparison with Equal Borda
- comparison with Existing Composite

These analyses serve different purposes. Criterion and weight perturbations test whether the ranking is overly dependent on one criterion or exact reliability-weight configuration. The γ analysis examines the strength of reliability differentiation. Fixed-versus-adaptive normalisation assesses whether VQ2 conclusions depend on scaling assumptions, while development-versus-held-out evaluation tests whether parameter refinement generalises beyond the evidence used for selection.

Comparisons with Equal Borda and the Existing Composite method provide reference points for determining whether RDRA's additional reliability-aware mechanism produces meaningful advantages or whether a simpler existing approach is sufficient.

Therefore, these checks strengthen the robustness of the methodology by reducing the risk that the final conclusions are an artefact of a single analytical choice.

8. Validity, Reliability and Reproducibility

8.1 Internal Validity

Internal validity is strengthened by ensuring that all aggregation methods are evaluated under identical experimental conditions. The same:

- candidate configurations
- deployment evidence
- criterion subsets
- corruption scenarios
- evaluation measures

are used when comparing RDRA with Equal Borda and the Existing Composite method.

This matched design reduces the risk that observed ranking differences are caused by inconsistent experimental conditions rather than the aggregation method itself. Development and held-out subsets are also separated during parameter refinement so that the γ parameter is not selected using the same evidence later used for validation. This helps reduce parameter-selection bias and supports a fairer assessment of generalisation.

8.2 Construct Validity

Construct validity depends on clearly distinguishing the concepts measured by the methodology. The research separates:

predictive performance $\neq$ criterion preference $\neq$ empirical reliability

Predictive performance represents how effectively a candidate performs its task, criterion preference represents the relative importance assigned to a deployment objective, and empirical reliability represents how reproducibly a criterion ranks the candidate configurations across repeated evaluation evidence.

RDRA specifically operationalises the third construct. This distinction is important because the reliability weights used by RDRA should not be interpreted as stakeholder preferences or subjective criterion importance. Clearly defining these constructs reduces the risk of drawing conclusions about one concept from evidence that measures another.

8.3 Statistical Conclusion Validity

Statistical conclusion validity is strengthened using multiple complementary statistical procedures, including:

- paired experimental comparisons
- 95% confidence intervals
- effect-size estimation
- multiple-comparison correction
- non-parametric robustness checks
- bootstrap uncertainty estimation.

The paired design is appropriate because competing aggregation methods are evaluated under the same perturbation conditions. Confidence intervals quantify uncertainty around estimated differences, while effect sizes indicate the magnitude of observed changes rather than relying only on statistical significance. Holm correction reduces the risk of false-positive findings across multiple VQ1 comparisons, and Wilcoxon signed-rank analysis provides a robustness check where paired differences depart from normality. Bootstrap analysis is used in VQ2 to estimate uncertainty around the difference in responsiveness between aggregation methods.

Therefore, these procedures decrease the likelihood that random variation is interpreted as meaningful methodological improvement.

8.4 External Validity

External validity is intentionally bounded by the nature of the experimental datasets. AI4I is a synthetic predictive-maintenance dataset, while C-MAPSS is based on simulated engine-degradation data. Consequently, the findings should not be interpreted as universal evidence that RDRA will outperform alternative methods in every industrial environment.

However, the use of two substantially different predictive-maintenance tasks strengthens the breadth of the evaluation. AI4I represents machine-failure classification, whereas C-MAPSS

represents sequential RUL prediction. This cross-task design provides broader evidence than evaluation on a single benchmark while still maintaining appropriate limits on generalisation.

Future validation using real industrial sensor data would be required before making stronger claims about operational deployment across different industries or equipment types.

8.5 Reproducibility

Reproducibility is supported through a documented and repeatable computational workflow. The project preserves:

- public benchmark datasets
- fixed random seeds
- documented preprocessing procedures
- repeatable notebooks
- explicit corruption protocols
- documented model configurations
- stored criterion-ranking and reliability tables
- preserved statistical outputs and figures

These measures allow the experimental workflow to be reconstructed and reduce dependence on undocumented manual decisions. Saving intermediate evidence also avoids unnecessary recomputation and makes it easier to trace the final RDRA rankings back to the underlying deployment scores and reliability estimates.

Reproducibility is particularly important in this research because the proposed reliability weights are themselves derived from repeated experimental evidence. Therefore, a repeatable workflow is necessary not only for reproducing the final ranking but also for verifying how the empirical reliability estimates were obtained.

9. Research Plan, Timeline and Milestones

9.1 Research Plan

Stage	Research activity	Principal output
1	Literature and problem refinement	Research gap and MRQ
2	Dataset preparation	AI4I and C-MAPSS pipelines
3	Candidate-model development	ML/DL configurations
4	Deployment-evidence generation	Multi-dimensional evidence
5	Criterion-ranking construction	Candidate rankings
6	RDRA development	Reliability-aware consensus
7	Criterion-uncertainty validation	VQ1 evidence
8	Data-quality and preference analysis	VQ2/VQ3 evidence

9	Statistical and sensitivity analysis	Comparative validation
10	Interpretation and writing	Final research report

9.2 Research Timeline

Activity	Period	Status
Initial literature review	SIT723	Completed
Initial research-question development	SIT723	Completed
Initial methodology design	SIT723	Completed
AI4I/C-MAPSS preparation	Earlier research phase	Completed
Baseline ML/DL experiments	Earlier research phase	Completed
Deployment-oriented evaluation	Earlier research phase	Completed
Initial MCDM/model-selection work	SIT746	Completed
Research-gap refinement	SIT746	Completed
Core RDRA implementation	SIT746	Completed
Reliability estimation	SIT746	Completed
VQ1 evaluation	SIT746	Completed
VQ1 statistical validation	SIT746	Completed
RDRA refinement	SIT746	Completed
VQ2 analysis	SIT746	Completed
VQ2 statistical validation	SIT746	Completed
VQ3 decision analysis	SIT746	Completed
C-MAPSS complementary analysis	SIT746	Completed
Expanded literature review	SIT746	Ongoing
Complete methodology draft	SIT746	Current
Integrated results/discussion	Remaining period	Planned
Supervisor review and revision	Before final submission	Planned

9.3 Key Milestones

Milestone	Completion evidence	Status
M1	Literature foundation	Completed
M2	Research problem and MRQ	Completed
M3	Dataset pipelines	Completed
M4	Candidate models	Completed
M5	Deployment framework	Completed
M6	Core RDRA implementation	Completed
M7	Baseline aggregation methods	Completed
M8	25 VQ1 perturbations	Completed
M9	VQ1 statistical validation	Completed
M10	RDRA refinement	Completed

M11	16 VQ2 scenarios	Completed
M12	VQ2 bootstrap validation	Completed
M13	VQ3 decision analysis	Completed
M14	C-MAPSS cross-task analysis	Completed
M15	Complete literature review	Current
M16	Complete methodology	Current
M17	Integrated results/discussion	Planned
M18	Supervisor revision cycle	Planned

10. Ethical Considerations and Research Risks

10.1 Human Participants, Consent and Privacy

The current research does not involve human participants, interviews, surveys, identifiable personal information, intervention with live industrial equipment.

Therefore, participant consent and confidentiality are not directly applicable. The VQ3 preference scenarios are computationally sampled rather than collected from real stakeholders.

Public benchmark data are used instead of confidential industrial records

10.2 Ethical Approval and Permissions

Formal human-research ethics approval is unlikely to be required under the present public-data and non-participant design. Any future extension involving participants, proprietary industrial data, identifiable employee information or commercially sensitive maintenance records would require appropriate approval before data collection.

10.3 Research Integrity

Because the research evaluates its own proposed method, results must be reported transparently, including positive findings, negative findings, non-significant differences, equivalence with baselines, cases where an existing method performs better and methodological limitations. Parameter-selection procedures must remain separated from held-out validation evidence.

10.4 Algorithmic and Operational Risk

PdM recommendations can affect maintenance decisions, potentially contributing to missed faults, unnecessary maintenance, financial loss or safety implications. The current study does not control live industrial equipment. Therefore, findings should be interpreted as methodological benchmark evidence rather than certification for safety-critical deployment.

10.5 Research Risk Register

Risk	Potential consequence	Mitigation
Benchmark-specific findings	Limited generalisability	Two predictive tasks and bounded claims
Synthetic/simulated data	Reduced industrial realism	Future real-world validation
Artificial corruption	Incomplete fault representation	Multiple corruption families and severities
Limited VQ1 perturbations	Statistical uncertainty	Paired design, CIs, effect sizes
Limited VQ2 scenarios	Uncertain responsiveness	Bootstrap analysis
Multiple comparisons	False-positive inflation	Holm correction [19]
Parameter-selection bias	Optimistic results	Development/held-out separation
Boundary γ result	Incomplete search	Development-only boundary sensitivity
Ranking ties	Misleading winner interpretation	Average ranks and tie-aware measures
Excessive stability	Failure to react to real changes	VQ2 responsiveness
Excessive sensitivity	Unstable recommendation	VQ1 consistency
Preference assumptions	Context-dependent conclusions	Multiple sampled scenarios
Operational assumptions	Misinterpreted cost claims	Explicit scenario labelling
Code errors	Invalid results	Version control and repeatable notebooks
Computational delay	Schedule disruption	Saved outputs and contingency time

11. Sustainability, Data Accessibility and Dissemination

11.1 Computational Sustainability

The project incorporates several computational sustainability measures to reduce redundant computation and improve efficiency. Predictions and deployment evidence are saved, intermediate outputs are reused, and unnecessary retraining is avoided. Parameter searches are restricted to justified ranges, and computational efficiency is evaluated as part of the deployment framework.

11.2 Long-Term Relevance

RDRA operates on criterion rankings rather than a specific predictive model, enabling long-term adaptability. The framework can accommodate new PdM datasets, alternative algorithms, additional deployment criteria, different robustness measures, and alternative empirical

reliability estimators. This ensures the methodological contribution remains relevant beyond the current experimental context.

11.3 Data and Research Accessibility

Data accessibility is supported through authoritative dataset references, documented preprocessing, fixed seeds, explicit experiment configurations, machine-readable result tables, and preserved statistical outputs. Derived results are stored in accessible formats such as CSV to facilitate reproducibility and future analysis.

11.4 Research Preservation

Research artefacts, including source code, notebooks, preprocessing definitions, model configurations, deployment scores, rankings, reliability weights, statistical outputs, figures, and software-environment information, are preserved using version control. This provides an auditable history of methodological development and supports long-term reproducibility.

11.5 Environmental Sustainability

Although no quantitative carbon-footprint estimate is provided, practical sustainability measures focus on reducing unnecessary experiments and avoiding excessive parameter searches. These measures help minimise computational resource usage.

11.6 Dissemination

Dissemination pathways include the final master’s research report, supervisor or research-group presentations, university seminars or posters, and potential future conference or journal submissions. Reproducible code and derived outputs may also be shared subject to licensing and institutional requirements.

12. Methodological Limitations and Conclusion

12.1 Limitations

Several limitations constrain the interpretation of the research.

First, AI4I is synthetic and C-MAPSS is simulator-derived.

Second, the corruption scenarios approximate selected forms of sensor degradation rather than all industrial faults.

Third, empirical criterion reliability depends on the candidate set, dataset, metrics, repetitions and experimental protocol.

Fourth, operational penalties and preference scenarios are analytical assumptions rather than directly measured industrial costs or stakeholder priorities.

Fifth, parameter refinement can create overfitting if development and validation data are not separated.

Finally, RDRA is not assumed to be universally superior. Its value is supported only to the extent demonstrated by the predefined experimental evidence.

12.2 Alignment Between Research Gap, Method and Validation

Research issue	Methodological response
Deployment objectives conflict	Criterion-level rankings
Measures have incompatible units	Normalised Borda transformation
Equal aggregation assumes equal influence	Reliability-aware weighting
Criterion rankings have unequal reproducibility	LOO Kendall stability estimation
Stability alone may be meaningless	Evidence-alignment requirement
Criteria may vary	VQ1 criterion uncertainty
Sensor evidence may degrade	VQ2 responsiveness
Deployment priorities may vary	VQ3 decision usefulness
Differences may occur by chance	Statistical validation
Results may depend on assumptions	Sensitivity and held-out analysis
Benchmark results may not generalise	Complementary C-MAPSS evidence and bounded claims

12.3 Conclusion

This methodology provides a quantitative, reproducible and falsifiable approach to deployment-oriented PdM model selection when multiple deployment criteria conflict and their supporting evidence has different empirical reproducibility.

The core RDRA method estimates criterion-ranking stability:

$$s_j = \frac{1}{R} \sum_{r=1}^R \max(0, \tau_{jr}),$$

constructs reliability weights:

$$w_j = \frac{s_j}{\sum_{k=1}^m s_k},$$

converts ranks to normalised Borda scores:

$$B_{ij} = \frac{n - r_{ij}}{n - 1},$$

and calculates:

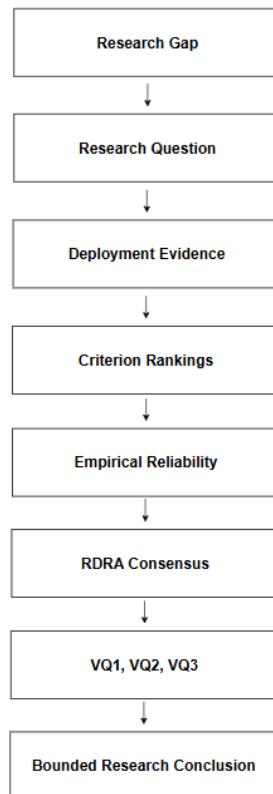
$$S_i = \sum_{j=1}^m w_j B_{ij} .$$

The method is then examined through three complementary perspectives:

- VQ1 = Consistency under Criterion Uncertainty
- VQ2 = Responsiveness under Data-Quality Uncertainty
- VQ3 = Decision Usefulness under Preference Uncertainty

C-MAPSS provides complementary responsiveness evidence from a different predictive task, while statistical and sensitivity analyses determine whether observed differences are robust rather than incidental.

The complete methodological pathway is:



This directly addresses the identified research gap by testing whether empirical criterion-ranking reproducibility can provide useful reliability information when conflicting deployment evidence must be aggregated into a defensible predictive-maintenance model recommendation.

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